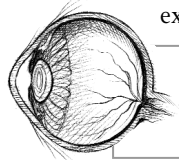


measuring this metric is as simple as taking a picture or analysing the images from a security video camera. A computer analyses the image to locate a 'face' within it, applies algorithms to compensate for factors such as lighting conditions and creates a normalised face image. This

image is then compared with an existing database.



- Extremely accurate with no false matches in more than 2 million.
- Intrusive, expensive, difficult to deploy and use.

One technique called the Facelt algorithm by Visionics works on the assumption that all faces have the same recurring elements that form it. These elements are small and many. The Facelt algorithm catalogues a series of such elements in your face, analysing their occurrence and geometric combinations. The company claims to achieve high levels of accuracy and produce quick search results.

Fingerprinting is a common and well-known biometric. You place your finger on a scanner, which uses either optical or electromagnetic means to take its snapshot. While optics may be obscured by dirt on the finger or stains on the scanner surface, electromagnetic sensors take that in their stride and have the

additional benefit of being able to detect the difference between live and dead fingers by their difference in capacitance. The software analyses your finger for patterns such as loops, whorls and arches. It takes the sampling of ten to fifteen points on your finger and stores that in an encrypted form. Authentication is a simple matter of comparing the finger you

offer with the finger signature stored in the database.

The retinal and iris patterns are quite

unique to individuals, far more so than any biometric yet devised. If you have ever been for 'computerised' eye testing, you have some idea of how this technology interfaces with users. You need to put your face in front of a camera, and align it carefully to a specific position. A low intensity Infrared beam shines through your eye, hitting the inside surface of the retina and warming it up. When warmed up, the blood vessel pattern shows up more clearly. This is also one of the reasons why movies like 'Demolition Man' are a little off-track—you can't rip off someone's eyeball and expect the system to work! The camera takes a snapshot of the retina or iris, and the digitised image is processed by the software. Again, only key points on the retina are stored into the database during enrolment. For authentication, your retinal pattern from these key areas is compared to that stored in the database.

Despite the high degree of reliability, iris and retina scans remain the domains of establishments with extreme security requirements. The scanning process is quite cumbersome and requires significant co-operation by the subject. It is also the most intrusive and works with the eye, an organ that most people are quite sensitive about.



- The only biometric that works over a phone.
- Flummoxed by illness, much easier to spoof than most other biometrics.

Putting it to use

Fingerprinting is quite convenient to deploy, and is now showing up on an experimental basis in many US supermarkets. Customers are invited to give their credit card information and other personal details to the store, and tie it in with their fingerprint data. The next time,

the customer can pay for purchases with just a fingerprint. Convenience is likely to be the main draw for public biometric systems, but with manufacturers also pushing the accuracy of biometrics, applications with higher security requirements are likely to emerge with ATMs being an obvious candidate.

Deploying fingerprinting in a corporate environment for either employee attendance or as a security measure is also a growing trend. One Indian manufacturer offers fingerprint terminals for just over Rs 60,000, but there are additional costs involved for either a dedicated computer or Rs 20,000 for a controller. Electronic locks that can be operated by such a device are sold for Rs 4,000 and you also need to factor in the cost of installation and training for all employees who must use the system.

Fingerprint scanners for personal computing devices are commonly available for \$100 to \$150 (between Rs 5,000 to Rs 7,500) with bundled software. Though simple pads like the Sony FIU-710 are common, they come in many different form factors. Some laptops, like Toshiba's 8200 series, can use PC Card fingerprint readers to provide BIOS level security. The Biolink U-Match Mouse has a thumb scanner located at the thumb rest area of the mouse.

The final word

Biometrics are going to be on a terminal near you soon enough, but should you blindly trust them? Not really. The propaganda on biometrics would have you believe that it is the definitive answer. Don't.

Security is not just about putting big locks on the front door, it also involves making sure all the windows are shut. As a security paranoiac, you will have to use multiple layers of security that include biometrics, smart cards and even the commonplace measure of passwords. Each one can be individually fooled or bypassed, but the combination makes

the system more secure as a whole. Biometric data must reside on the same kind of servers and networks that are hacked every day. If credit card numbers can be stolen, why not biometric signatures?

Yet, no matter what the security issues, biometrics systems do make ultra cool gizmos. ■

SUMOD HAJELA

Online Resources

Sans Institute Reading Room

A selection of articles explaining how biometrics works and how it can be integrated into a security framework.

<http://rr.sans.org/>

Biometric Consortium

A US Government site dedicated to biometrics and its applications.

www.biometrics.org/html/examples.html

BioPrivacy Initiative

Analysis of privacy risk factors from biometrics, and guidelines on proper usage of the technology.

www.bioprivacy.org

ACLU

A site dedicated to civil liberty. While it is completely American in focus, it deals with issues we will face with public deployment and over-reliance on biometrics in the coming years.

<http://aclu.org/>